CCE-321: Computer Peripheral and Interfacing

Important Question Analysis — Based on 11 Past Papers (2014–2023 Sessions)

Papers analyzed: Midterms (2021, 2023) and Final Exams from sessions 2011-12, 2012-13, 2013-14, 2015-16, 2017-18, 2018-19, 2019-20, 2020-21, plus one undated final. Questions repeated across multiple exam years, or carrying high marks consistently, are marked as important.

Legend

★★★ High Priority	Topic/question appeared in 4 or more separate exams — very likely to repeat.
★★ Medium Priority	Topic/question appeared in 2–3 exams — reasonably likely to repeat.
★ Worth Reviewing	Appeared once but carries high marks or is a core syllabus topic.

Topic Frequency Summary (Ranked)

Priority	Topic	Times Seen	Typical Marks
★★★	RAM/ROM Interfacing with Microprocessor (address maps, chip select, memory decoding)	6+	5–9
★★★	LVDT / RVDT — construction & step-by-step working	5	5–6
★★★	Sensors vs Transducers — differences & definitions	5	3–4
★★★	8255 PPI — pin configuration, operation, control word	5	4–6
★★★	CRT Display — delta-delta shadow mask, raster vs random scan, true color	5	3–6
★★★	IoT — components, communication models, security, dimensions	5	3–6
★★★	Keyboard Interfacing — key debouncing, make/break code, software interfacing flowchart	5	2–5
★★★	8279 Programmable Keyboard/Display Controller — architecture & working	4	5–6
★★★	Printers — impact vs non-impact, inkjet, laser working principle	4	4–6
★★★	USB — definition, functions, topology, packet types	4	3–5
★★	IEEE-754 32-bit Floating Point Conversion	3	3–4
★★	8251A PCI — functional block diagram & sections	3	3–7
★★	RS-232 / Asynchronous Serial Communication & Handshaking	3	5–8
★★	ASCII vs EBCDIC	2	2
★★	Overflow Flag — when set/reset, examples	3	2–3
★★	FDC Data Read/Write + FM/MFM/RLL Encoding (identical Q repeated)	2	14 (5+9)
★★	I/O Types — programmed, interrupt-initiated, device-initiated / DMA	4	2–4
★★	HCI & Norman's (Seven) Principles / Usability	3	3–5
★★	RAID / SCSI / ATA / IDE storage interfaces	2	3–7
★★	Buffers, Spoolers, Interfaces — definitions & characteristics	3	2–6
★	Wireless Sensor Networks — MAC protocols (T-MAC/S-MAC, ALOHA, CSMA hidden/exposed terminal)	1	3–4
★	Real-Time Systems — periodic/aperiodic task models, embedded vs real-time	2	3–4
★	Detailed Sensors — piezoelectric, Hall effect, thermistor, pyrometer, thermocouple, manometer	1	2–6
★	Maskable vs Nonmaskable Interrupts / Software Interrupt	2	2–3

Compiled Question Bank (Grouped by Topic)

★ marks next to a question indicate it (or a close variant) appeared in more than one exam paper. Years shown refer to the exam session/year printed on the paper.

1. Memory (RAM/ROM) Interfacing with Microprocessor ★★★

★ Interface two chips of 4k RAM and 1/2 chip(s) of 2k ROM with processor for a given starting address (8000H / 9000H / A000H). *[Midterm 2021, Midterm 2023, Final 2023]*

★ Connect the microprocessor to a given ROM/RAM chip (with A0–A9, CS, RD/WR lines shown) to realize a memory map (e.g. 0000H–03FFH or 0000H–0FFFH). Draw a neat logic diagram and analyze the memory map. *[Final 2018, Final 2023]*

★ Illustrate an 8-bit microprocessor interfaced to a 4K RAM system using full/partial decoding; show the address map. *[Final 2016]*

• Design a 1K x 8 RAM system using a 512 x 8 RAM chip as the building block (CS1=L, CS2=H). Draw logic diagram; determine memory map in hexadecimal using linear decoding with don't-care conditions as 1's. *[Final 2023]*

• Given a memory with a 14-bit address and 8-bit word size: how many bytes can be stored? If built from 1K x 1 RAM chips, how many are required? How many bits for chip select? *[Final 2023]*

• Suppose a processor has 16-bit address lines and is byte addressable. Interface 4096x8 RAM with the processor — memory capacity, address lines, number of RAM chips, size of address decoder. *[Final 2021]*

• What are the basic differences between standard I/O and memory-mapped I/O? *[Final 2023 Midterm]*

• Differentiate between ISA bus and PCI bus. *[Final 2016]*

2. LVDT / RVDT ★★★

★ What is an LVDT (or RVDT)? How does it work step by step? Explain with construction diagram. *[Final 2020 (RVDT), Peripheral, Final 2018 (Helipots & LVDT), Final 2023, Final 2022]*

• Describe the effect of variation of dielectric constant for measurement of displacement in a capacitive transducer. *[Final 2018]*

★ How does a capacitive sensor operate to measure linear displacement? Typical applications where non-contact sensors are preferred; explain with diagram. *[Final 2023, Final 2022]*

3. Sensors, Transducers & Transducer Theory ★★★

★ What is the main difference between Sensors and Transducers/Transforms? *[Peripheral, Final 2020, Final 2021]*

• Define transducers. Write down the advantages of electrical signal as output of a transducer. *[Final 2018]*

• What are the definitions and characteristics of active and passive transducers? How do sensors and transducers differ in function and application? *[Final 2022]*

★ What do you mean by nonlinearity error? How is it different from Hysteresis error? *[Final 2020, Final 2023, Final 2022]*

• Define piezo-electric effect. Derive the equation of voltage sensitivity due to piezo-electric effect. *[Final 2018]*

• Write short notes on: photoemissive cell, photoconductive cell, photovoltaic cell. *[Final 2018]*

★ What is Hall effect? Describe the working principle of Hall effect transducers/sensors. *[Final 2018, Final 2023]*

• Write down applications and limitations of thermistors. *[Final 2018]*

• Define pyrometer. Describe the working principle of radiation pyrometer with advantages and limitations. *[Final 2018]*

• What is thermocouple? Briefly discuss construction and operating principle. *[Final 2018]*

• Write short notes on U-type manometer and Well-type manometer. *[Final 2018]*

• 'Bellows are more sensitive than capsules in Fluid pressure sensor' — True or False? Justify. *[Final 2023]*

4. 8255 / 82C55 Programmable Peripheral Interface (PPI) ★★★

★ Show the pin configuration of 8255 PPI and describe the operation performed by each unit separately. *[Final 2020, Final 2022]*

• Define 8155 PPI. Write down its features. *[Peripheral]*

• What is 82C55 PPI? What will be the operation of a stepper motor connected with 82C55 PPI? *[Peripheral]*

★ Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output, port-B as input, port-C as output. *[Final 2018-19, Final 2023]*

• How can we assign input and output ports of 8255? *[Final 2022]*

5. Display Systems — CRT, LCD, Color, Scan Types ★★★

★ Explain the operation of a delta-delta shadow-mask CRT. *[Final 2018, Final 2016]*

★ Define raster scan display and random scan display. What are the differences between them? *[Final 2015, Final 2014, Final 2022]*

★ What do you mean by true color representation? Does it represent any disadvantages / how to handle problems? If we use N bits per primary color, how many simultaneous colors possible? *[Final 2018, Final 2018-19]*

• Write short notes on: Interlaced refresh procedure, 10H8, Bitmap and Pixmap. *[Final 2018]*

★ What are the advantages and disadvantages of LCD over CRT display / LCD panel over conventional CRT monitor? *[Final 2016, Final 2015, Final 2014]*

• What determines the quality of a monitor? / Write down the selection criteria of a monitor. *[Final 2020, Final 2015]*

• Explain in detail the working principle of an LCD monitor and how its components interact to produce images. *[Final 2022]*

★ Serialize the steps associated in turning on or off a pixel in an LCD monitor. *[Final 2023, Final 2021]*

• What is a plasma monitor? *[Final 2020]*

• Find CMY coordinates of a color given RGB values; direct coding with N bits per color — how many possible colors per pixel; pitch of a color CRT; why color printers use black pigment. *[Final 2022]*

• If we want to cut a 512x512 sub-image from an 800x600 image, find pixel coordinates of the lower-left corner. *[Final 2022]*

6. Internet of Things (IoT) ★★★

★ What are the different components of IoT? How does IoT affect our everyday lives? *[Final 2018-19, Final 2019-20]*

★ What is a communication model in IoT? Explain different types of communication models with diagram and example / impact on data transmission, scalability, energy efficiency. *[Final 2018-19, Midterm 2023]*

• What does it mean by Internet of Underground Things and Internet of Underwater Things? *[Midterm 2023]*

• The IoT is often described in three key dimensions. Explain each with examples; how do they enable a seamless IoT ecosystem? *[Midterm 2023]*

★ What are the different types of IoT security measures/best practices for securing IoT devices? *[Final 2019-20, Final 2018-19]*

★ Briefly explain the most important communication protocols used by IoT (e.g. MQTT, M2M). *[Final 2018-19, Final 2023]*

• What are the key components and characteristics of a digital ecosystem? Challenges/opportunities in managing one. *[Final 2019-20]*

7. Keyboards & Interfacing ★★★

★ What is key debouncing? Why is it needed? *[Final 2016, Final 2022]*

• Define make code, break code, typematic, persistence, aspect ratio and resolution. *[Final 2016]*

★ When a key on a keyboard/computer is pressed, a switch is activated to generate signals. Give a brief description of four such switching methods. *[Final 2016, Final 2020]*

★ Show the software keyboard interfacing process through a flow chart. Describe each step for a 4x4 keyboard. *[Final 2020, Final 2018-19, Final 2021]*

• Modern keyboards support bidirectional data transfer. How? *[Final 2016]*

8. 8279 Programmable Keyboard/Display Controller ★★★

★ Draw the architectural diagram of 8279 keyboard/display controller. How does 8279 keyboard work? *[Final 2020, Final 2023]*

• Categorize and explain the operational modes of 8279. How many ways is the keyboard interfaced with the CPU? *[Final 2018-19]*

★ Differentiate between programmed I/O and interrupt-initiated I/O. *[Final 2020, Final 2018 (device-initiated I/O)]*

9. Printers ★★★

★ What is Impact Printer? How does it work? Differentiate impact and non-impact printers. *[Final 2020, Peripheral]*

• How does an inkjet printer work? Explain with figure. *[Peripheral]*

★ How does a laser printer work? Explain with functional diagram. *[Peripheral, Final 2022]*

• Write down the steps involved in starting a printing sequence using a printer controller chip. *[Final 2016]*

★ An interface incorporates several components including communication type and interface software. List eight major communication types in case of a printer. *[Final 2023, Final 2018-19, Final 2021]*

10. USB & Serial/Parallel Buses ★★★

★ Define USB. What are the functions of USB ports? Write down the limitations of USB ports. *[Final 2015, Final 2014]*

• Differentiate between IEEE1394 and USB. *[Final 2015]*

★ Sketch the three-tier topology of USB. Write down the benefits of the three-tier topology. *[Final 2015, Final 2014]*

• In USB communication, a transaction involves up to three packets. Name these packets and describe the service each provides. *[Final 2023]*

• How do USB peripherals communicate with the microprocessor? Give the pin-out for USB Type C 3.0. *[Final 2018-19]*

★ What is PCI Express? Write down features and applications of PCI Express bus. *[Final 2015, Final 2014]*

★ Differentiate between serial ATA and parallel ATA. Why is serial ATA used instead of parallel ATA? *[Final 2015, Final 2014, Final 2016]*

• Differentiate between ISA, VESA and EISA buses / compare VESA and PCI buses. *[Final 2015 / Final 2014]*

★ Write short notes on AGP, Plug and Play, SCSI. *[Final 2015, Final 2014]*

★ Differentiate between IRQ and DMA channels. *[Final 2015, Final 2014]*

11. Number Systems & Floating Point ★★

★ Convert a given decimal number (e.g. 39887.5625, -1313.3125) to IEEE 32-bit floating point format. *[Final 2020, Final 2022]*

• Interpret a given hex value (e.g. EFE16) as unsigned, signed, and floating point numbers that a microprocessor can understand. *[Final 2018-19]*

• How can signed binary numbers be represented? What is the range of signed values stored in 20 bits? *[Final 2020]*

• Perform binary addition of two signed decimal numbers; determine carry, sign, overflow, and zero flags. *[Final 2018]*

★ When is an overflow flag set or reset? Give examples. *[Final 2020, Final 2016, Final 2018]*

12. Serial Communication / RS-232 / 8251A ★★

★ Draw the functional block diagram of 8251A PCI. Briefly explain each section. *[Final 2020, Peripheral]*

★ What is RS-232 Handshaking / asynchronous communication protocol? Briefly describe RS-232C connection with a simple example. *[Final 2020, Peripheral]*

• Assume ASCII character 'A' is framed between start bit and two stop bits, LSB sent first — define character-oriented transmission, how data is packed, bits per character, when data begins/ends, sync vs async, best method, framing diagram. *[Final 2020]*

• Which process of communication is controlled by an external circuit, used for data transfer between memory and I/O? Draw block diagram of a typical DMA controller. *[Peripheral]*

• What are the basic principles of the two-wire handshaking method of data transfer? *[Final 2020]*

• What is the IEEE 488 bus system? Explain its operation based on handshaking protocol with pin configuration. *[Peripheral]*

13. Coding & Interrupts ★★

★ What are the basic differences between ASCII and EBCDIC? *[Final 2018, Final 2016]*

• Define software interrupt with example. *[Midterm 2021]*

• What are the main types of interrupt? Describe with example. *[Midterm 2023]*

• Summarize the basic difference between maskable and nonmaskable interrupts. Describe how power failure interrupt is normally handled. *[Final 2023]*

• Define the three types of I/O. Identify each as microprocessor-initiated or device-initiated. *[Final 2023]*

• Describe the process of a device-initiated I/O transfer. *[Final 2016]*

14. Floppy Disk Controller & Encoding ★★

★ Describe Data Read and Data Write operations through FDC. *[Final 2018, Final 2016]*

★ Encode data 101000100110 using FM, MFM and RLL encoding formats. Show the waveforms. *[Final 2018, Final 2016]*

• What do you mean by 'CHRN' value? Describe IDE command execution protocols. *[Final 2016]*

15. Human Computer Interaction (HCI) ★★

• What is Usability in HCI? Why is usability important? Give an example. *[Midterm 2021]*

★ Write down the Norman's Seven Principles for HCI/IoT design. *[Midterm 2021, Final 2018-19, Final 2019-20]*

★ What is Human Computer Interaction? Briefly explain the most important components of HCI. *[Final 2018-19, Final 2019-20]*

16. Storage Interfaces — RAID, SCSI, ATA ★★

★ Define RAID. Describe the different levels of RAID. *[Final 2015, Final 2014]*

★ Differentiate between IDE/ATA and SCSI. Write down the advantages of SCSI over ATA. *[Final 2015, Final 2014]*

• Differentiate between ATA RAID and SCSI RAID. *[Final 2015]*

• What do you mean by solid state storage device? Write down features and advantages. *[Final 2015, Final 2014]*

17. Buffers, Spoolers & Interfaces ★★

★ What is buffer? Why do we use buffers? Differentiate between buffers and spoolers. *[Final 2015, Final 2014, Final 2018-19]*

• Define interfaces. What are the characteristics of computer peripherals? *[Final 2015, Final 2014]*

• Mention/list the main functions of an interface. *[Final 2019-20]*

• List the methods used in main memory array design. What are the advantages and disadvantages of each? *[Final 2023, Final 2021 (linear decoding problems)]*

18. Wireless Sensor Networks, Real-Time Systems & Misc. (Newer Syllabus) ★

• Which design features are essential in WSNs to optimize data collection, communication, and energy efficiency? *[Final 2023]*

• Write the primary objective/idea of Timeout-MAC (T-MAC). How does T-MAC differ from S-MAC in energy efficiency? *[Final 2023]*

• Write down Hidden Terminal Problem and Exposed Terminal Problem for CSMA MAC-Protocol in WSN. Explain with figure. *[Final 2023]*

• How does Slotted ALOHA improve upon basic ALOHA? Explain with frame transition diagrams. *[Final 2023]*

• What is the periodic and aperiodic task model in real-time systems? Explain with example. *[Final 2018-19]*

• Write down the difference between Embedded and Real-Time Systems. *[Final 2018-19]*

• Imagine designing a real-time system for a smart city — how would you incorporate periodic/aperiodic task models to manage operations? *[Final 2019-20]*

• Define proximity switches. Sketch configurations of contact-type proximity switches used in manufacturing automation; two applications of 'Reed switch'. *[Final 2023]*

• Explain the characteristics of RTS (Real-Time Systems). *[Final 2018-19]*

• What are the common problems with scanners? Share tips to troubleshoot basic scanner-related problems. *[Final 2018-19]*

• How does a paper scanner work? Explain with functional diagram. *[Peripheral]*

• What is the example of biometric technology? *[Peripheral]*

• Illustrate the Gantt chart for your project. (CCE-322 Sessional/Arduino specific questions on IOREF pin, Arduino IDE, ESP8266 WiFi module, digitalRead/digitalWrite tracing). *[CCE-322 Sessional 2021]*

Study Priority Recommendation: Focus first on the ★★★ topics (memory interfacing, LVDT/RVDT, sensors vs transducers, 8255 PPI, CRT/display systems, IoT, keyboard interfacing, 8279, printers, and USB) as they account for the large majority of marks across the last decade of exams. Then cover the ★★ topics for a well-rounded preparation, and skim the ★ items for completeness since the syllabus has shifted toward IoT/WSN topics in the most recent (2022–2023) papers.